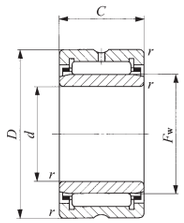
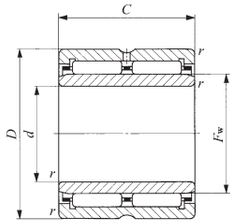
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring

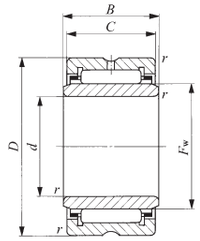




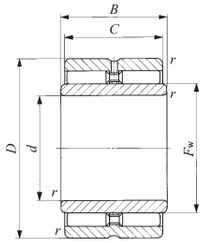
NA49 TAFI



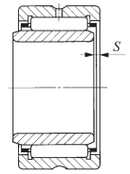
NA69

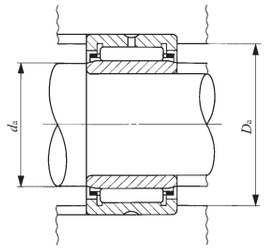


TRI



GTRI





Shaft dia. 75 — 90mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d
75	—	—	—	**TAFI 7510525**	—	—	675	75
	NA 4915	—	—	—	—	—	810	75
	—	—	—	**TAFI 7510535**	—	—	945	75
	—	**NA 6915**	—	—	—	—	1 480	75
	—	—	—	—	**TRI 7510845**	—	1 340	75
	—	—	—	—	—	**GTRI 7510845**	1 440	75
80	—	—	—	**TAFI 8011025**	—	—	710	80
	NA 4916	—	—	—	—	—	855	80
	—	—	—	**TAFI 8011035**	—	—	995	80
	—	**NA 6916**	—	—	—	—	1 560	80
85	—	—	—	**TAFI 8511526**	—	—	775	85
	—	—	—	**TAFI 8511536**	—	—	1 080	85
	NA 4917	—	—	—	—	—	1 280	85
	—	**NA 6917**	—	—	—	—	2 340	85
	—	—	—	—	**TRI 8511850**	—	1 640	85
	—	—	—	—	**TRI 8512045**	—	1 610	85
	—	—	—	—	—	**GTRI 8511850**	1 780	85
	—	—	—	—	—	**GTRI 8512045**	1 720	85
90	—	—	—	**TAFI 9012026**	—	—	820	90
	—	—	—	**TAFI 9012036**	—	—	1 140	90
	NA 4918	—	—	—	—	—	1 350	90
	—	—	—	—	**TRI 9012550**	—	1 870	90
	—	**NA 6918**	—	—	—	—	2 460	90
	—	—	—	—	—	**GTRI 9012550**	2 020	90

Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (3) rpm	Assembled inner ring
105	25	—	1	85	0.5	80	83	100	76 300	145 000	4 500	**LRT 758525**
105	30	—	1	85	1.5	80	83	100	86 200	169 000	4 500	**LRT 758530**
105	35	—	1	85	1.5	80	83	100	102 000	210 000	4 500	**LRT 758535**
105	54	—	1	85	1	80	83	100	138 000	331 000	4 500	**LRT 758554**
108	45	45.5	1	83	2.5	80	81	103	146 000	270 000	5 000	**LRT 758345**
108	45	45.5	1	83	—	80	81	103	190 000	396 000	1 800	**LRTZ 758345**
110	25	—	1	90	0.5	85	88	105	77 300	150 000	4 500	**LRT 809025**
110	30	—	1	90	1.5	85	88	105	87 300	175 000	4 500	**LRT 809030**
110	35	—	1	90	1.5	85	88	105	103 000	217 000	4 500	**LRT 809035**
110	54	—	1	90	1	85	88	105	143 000	351 000	4 500	**LRT 809054**
115	26	—	1	95	1	90	93	110	79 700	159 000	4 000	**LRT 859526**
115	36	—	1	95	2	90	93	110	106 000	231 000	4 000	**LRT 859536**
120	35	—	1.1	100	1	91.5	98	113.5	110 000	244 000	4 000	**LRT 8510035**
120	63	—	1.1	100	0.5	91.5	98	113.5	173 000	467 000	4 000	**LRT 8510063**
118	50	50.5	1	93	3	90	91	113	165 000	329 000	4 500	**LRT 859350**
120	45	45.5	1.5	95	2.5	93	93.5	112	155 000	305 000	4 000	**LRT 859545**
118	50	50.5	1	93	—	90	91	113	224 000	509 000	1 600	**LRTZ 859350**
120	45	45.5	1.5	95	—	93	93.5	112	204 000	455 000	1 600	**LRTZ 859545**
120	26	—	1	100	1	95	98	115	82 400	168 000	4 000	**LRT 9010026**
120	36	—	1	100	2	95	98	115	110 000	244 000	4 000	**LRT 9010036**
125	35	—	1.1	105	1	96.5	103	118.5	113 000	258 000	4 000	**LRT 9010535**
125	50	50.5	1.5	100	3	98	98.5	117	172 000	355 000	4 000	**LRT 9010050**
125	63	—	1.1	105	0.5	96.5	103	118.5	178 000	490 000	4 000	**LRT 9010563**
125	50	50.5	1.5	100	—	98	98.5	117	234 000	549 000	1 500	**LRTZ 9010050**

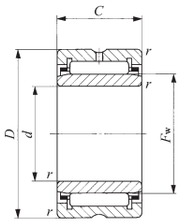
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable axial shift amount of inner ring to outer ring
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

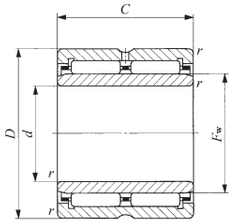
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring

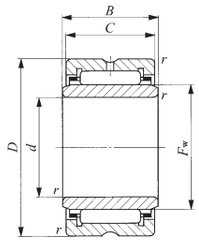




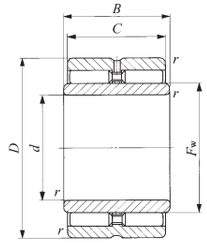
NA49 TAFI
NA48



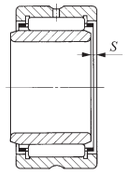
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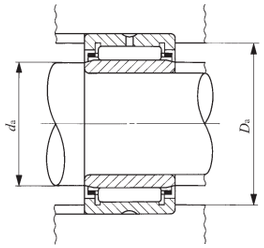


TRI



GTRI





Shaft dia. 95 — 150mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d
95	—	—	—	**TAFI 9512526**	—	—	860	95
	—	—	—	**TAFI 9512536**	—	—	1 190	95
	NA 4919	—	—	—	—	—	1 420	95
	—	**NA 6919**	—	—	—	—	2 580	95
100	—	—	—	**TAFI 10013030**	—	—	1 040	100
	—	—	—	**TAFI 10013040**	—	—	1 380	100
	—	—	—	—	**TRI 10013550**	—	2 040	100
	NA 4920	—	—	—	—	—	1 960	100
	—	—	—	—	—	**GTRI 10013550**	2 200	100
105	—	—	—	—	**TRI 10515350**	—	3 020	105
	—	—	—	—	—	**GTRI 10515350**	3 270	105
110	—	—	**NA 4822**	—	—	—	1 200	110
	NA 4922	—	—	—	—	—	2 120	110
120	—	—	**NA 4824**	—	—	—	1 300	120
	NA 4924	—	—	—	—	—	2 960	120
125	—	—	—	—	**TRI 12517860**	—	4 780	125
	—	—	—	—	—	**GTRI 12517860**	5 180	125
130	—	—	**NA 4826**	—	—	—	1 960	130
	NA 4926	—	—	—	—	—	4 030	130
135	—	—	—	—	**TRI 13518860**	—	5 100	135
	—	—	—	—	—	**GTRI 13518860**	5 530	135
140	—	—	**NA 4828**	—	—	—	2 100	140
	NA 4928	—	—	—	—	—	4 290	140
150	—	—	**NA 4830**	—	—	—	2 880	150
	NA 4930	—	—	—	—	—	6 380	150

Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(3) rpm	Assembled inner ring
125	26	—	1	105	1	100	103	120	84 700	178 000	4 000	**LRT 9510526**
125	36	—	1	105	2	100	103	120	113 000	258 000	4 000	**LRT 9510536**
130	35	—	1.1	110	1	101.5	108	123.5	116 000	271 000	3 500	**LRT 9511035**
130	63	—	1.1	110	0.5	101.5	108	123.5	182 000	514 000	3 500	**LRT 9511063**
130	30	—	1	110	0.5	105	108	125	106 000	240 000	3 500	**LRT 10011030**
130	40	—	1	110	1.5	105	108	125	134 000	324 000	3 500	**LRT 10011040**
135	50	50.5	1.5	110	3	108	108.5	127	183 000	395 000	3 500	**LRT 10011050**
140	40	—	1.1	115	1	106.5	113	133.5	145 000	329 000	3 500	**LRT 10011540**
135	50	50.5	1.5	110	—	108	108.5	127	245 000	603 000	1 400	**LRTZ 10011050**
153	50	50.5	1.5	115	3	113	113.5	145	233 000	414 000	3 500	**LRT 10511550**
153	50	50.5	1.5	115	—	113	113.5	145	315 000	614 000	1 300	**LRTZ 10511550**
140	30	—	1	120	1	115	118	135	93 200	239 000	3 500	**LRT 11012030**
150	40	—	1.1	125	1	116.5	123	143.5	152 000	357 000	3 000	**LRT 11012540**
150	30	—	1	130	1	125	128	145	96 900	259 000	3 000	**LRT 12013030**
165	45	—	1.1	135	2	126.5	133	158.5	187 000	435 000	3 000	**LRT 12013545**
178	60	60.5	1.5	140	2.5	133	138	170	307 000	625 000	3 000	**LRT 12514060**
178	60	60.5	1.5	140	—	133	138	170	409 000	923 000	1 100	**LRTZ 12514060**
165	35	—	1.1	145	1	136.5	143	158.5	116 000	340 000	3 000	**LRT 13014535**
180	50	—	1.5	150	2.5	138	148	172	215 000	540 000	2 500	**LRT 13015050**
188	60	60.5	1.5	150	2.5	143	148	180	320 000	675 000	2 500	**LRT 13515060**
188	60	60.5	1.5	150	—	143	148	180	423 000	989 000	1 000	**LRTZ 13515060**
175	35	—	1.1	155	1	146.5	153	168.5	120 000	363 000	2 500	**LRT 14015535**
190	50	—	1.5	160	2.5	148	158	182	224 000	580 000	2 500	**LRT 14016050**
190	40	—	1.1	165	1.5	156.5	163	183.5	168 000	446 000	2 500	**LRT 15016540**
210	60	—	2	170	3	159	168	201	324 000	712 000	2 500	**LRT 15017060**

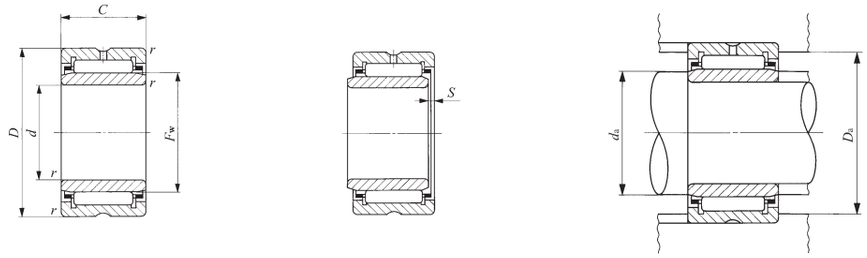
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable axial shift amount of inner ring to outer ring
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring





NA49 NA48

Shaft dia. 160 — 340mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d
160	—	—	**NA 4832**	—	—	—	3 050	160
	NA 4932	—	—	—	—	—	6 750	160
170	—	—	**NA 4834**	—	—	—	4 120	170
	NA 4934	—	—	—	—	—	7 110	170
180	—	—	**NA 4836**	—	—	—	4 340	180
	NA 4936	—	—	—	—	—	10 200	180
190	—	—	**NA 4838**	—	—	—	5 760	190
	NA 4938	—	—	—	—	—	10 700	190
200	—	—	**NA 4840**	—	—	—	6 040	200
	NA 4940	—	—	—	—	—	15 400	200
220	—	—	**NA 4844**	—	—	—	6 570	220
	NA 4944	—	—	—	—	—	16 700	220
240	—	—	**NA 4848**	—	—	—	10 200	240
	NA 4948	—	—	—	—	—	18 000	240
260	—	—	**NA 4852**	—	—	—	11 000	260
	NA 4952	—	—	—	—	—	31 100	260
280	—	—	**NA 4856**	—	—	—	15 800	280
	NA 4956	—	—	—	—	—	33 100	280
300	—	—	**NA 4860**	—	—	—	22 300	300
	NA 4960	—	—	—	—	—	51 400	300
320	—	—	**NA 4864**	—	—	—	23 700	320
	NA 4964	—	—	—	—	—	54 400	320
340	—	—	**NA 4868**	—	—	—	25 000	340
	NA 4968	—	—	—	—	—	57 300	340

Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(3) rpm	Assembled inner ring
200	40	—	1.1	175	1.5	166.5	173	193.5	173 000	474 000	2 500	**LRT 16017540**
220	60	—	2	180	3	169	178	211	337 000	761 000	1 900	**LRT 16018060**
215	45	—	1.1	185	1.5	176.5	183	208.5	211 000	567 000	1 900	**LRT 17018545**
230	60	—	2	190	3	179	188	221	347 000	810 000	1 900	**LRT 17019060**
225	45	—	1.1	195	1.5	186.5	193	218.5	218 000	602 000	1 900	**LRT 18019545**
250	69	—	2	205	3	189	203	241	434 000	989 000	1 900	**LRT 18020569**
240	50	—	1.5	210	1.5	198	208	232	249 000	726 000	1 800	**LRT 19021050**
260	69	—	2	215	3	199	213	251	440 000	1 020 000	1 700	**LRT 19021569**
250	50	—	1.5	220	1.5	208	218	242	255 000	766 000	1 600	**LRT 20022050**
280	80	—	2.1	225	4	211	223	269	518 000	1 120 000	1 600	**LRT 20022580**
270	50	—	1.5	240	1.5	228	238	262	266 000	833 000	1 500	**LRT 22024050**
300	80	—	2.1	245	4	231	243	289	536 000	1 200 000	1 400	**LRT 22024580**
300	60	—	2	265	2	249	262	291	345 000	1 150 000	1 300	**LRT 24026560**
320	80	—	2.1	265	4	251	262	309	565 000	1 320 000	1 300	**LRT 24026580**
320	60	—	2	285	2	269	282	311	354 000	1 220 000	1 100	**LRT 26028560**
360	100	—	2.1	290	4	271	287	349	847 000	1 900 000	1 100	**LRT 260290100**
350	69	—	2	305	2.5	289	302	341	486 000	1 550 000	950	**LRT 28030569**
380	100	—	2.1	310	4	291	307	369	877 000	2 040 000	950	**LRT 280310100**
380	80	—	2.1	330	2.5	311	327	369	610 000	1 900 000	900	**LRT 30033080**
420	118	—	3	340	4	313	337	407	1 130 000	2 650 000	850	**LRT 300340118**
400	80	—	2.1	350	2.5	331	347	389	635 000	2 040 000	750	**LRT 32035080**
440	118	—	3	360	4	333	357	427	1 170 000	2 830 000	750	**LRT 320360118**
420	80	—	2.1	370	2.5	351	367	409	651 000	2 140 000	700	**LRT 34037080**
460	118	—	3	380	4	353	377	447	1 220 000	3 020 000	700	**LRT 340380118**

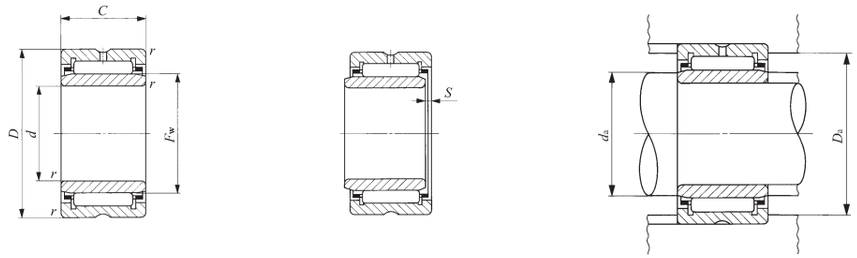
Notes(1) Minimum allowable value of chamfer dimension *r*
(2) Allowable axial shift amount of inner ring to outer ring
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring





NA49 NA48

Shaft dia. 360 — 440mm

Shaft dia. mm	Identification number NA 49	NA 69	NA 48	TAFI	TRI	GTRI	Mass (Ref.) g	d	Boundary dimensions mm D	C	B	$r_{s\,min}$ (1)	F_w	S (2)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (3) rpm	Assembled inner ring
360	—	—	**NA 4872**	—	—	—	26 400	360	440	80	—	2.1	390	2.5	371	387	429	680 000	2 320 000	650	**LRT 36039080**
	NA 4972	—	—	—	—	—	60 200	360	480	118	—	3	400	4	373	397	467	1 260 000	3 200 000	600	**LRT 360400118**
380	—	—	**NA 4876**	—	—	—	44 600	380	480	100	—	2.1	415	3	391	412	469	951 000	2 860 000	600	**LRT 380415100**
	NA 4976	—	—	—	—	—	90 300	380	520	140	—	4	430	5	396	427	504	1 540 000	4 030 000	500	**LRT 380430140**
400	**NA 4980**	—	—	—	—	—	94 400	400	540	140	—	4	450	5	416	447	524	1 590 000	4 270 000	500	**LRT 400450140**
420	**NA 4984**	—	—	—	—	—	98 500	420	560	140	—	4	470	5	436	467	544	1 640 000	4 510 000	500	**LRT 420470140**
440	**NA 4988**	—	—	—	—	—	131 000	440	600	160	—	4	490	5	456	487	584	1 910 000	5 140 000	400	**LRT 440490160**

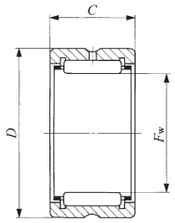
Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable axial shift amount of inner ring to outer ring
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

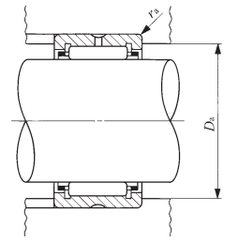
MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring, Inch Series





BR



Shaft dia. 15.875 — 47.625mm

Shaft dia. mm (inch)	Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) F_w	D	C	Standard mounting dimensions mm D_a Max.	$r_{as\,max}$ (1)	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
15.875 (5/8)	**BR 101812**	49	15.875 (5/8)	28.575 (1 1/8)	19.050 (3/4)	24.5	0.6	18 900	19 700	25 000
19.050 (3/4)	**BR 122012**	56	19.050 (3/4)	31.750 (1 1/4)	19.050 (3/4)	26.5	1	21 700	24 400	20 000
	BR 122016	75	19.050 (3/4)	31.750 (1 1/4)	25.400 (1)	26.5	1	27 600	33 100	20 000
22.225 (7/8)	**BR 142212**	63	22.225 (7/8)	34.925 (1 3/8)	19.050 (3/4)	29.7	1	23 000	27 100	18 000
	BR 142216	84.5	22.225 (7/8)	34.925 (1 3/8)	25.400 (1)	29.7	1	29 100	36 800	18 000
25.400 (1)	**BR 162412**	69	25.400 (1)	38.100 (1 1/2)	19.050 (3/4)	32.9	1	25 300	31 900	16 000
	BR 162416	92.5	25.400 (1)	38.100 (1 1/2)	25.400 (1)	32.9	1	32 100	43 300	16 000
28.575 (1 1/8)	**BR 182616**	102	28.575 (1 1/8)	41.275 (1 5/8)	25.400 (1)	36	1	34 900	49 900	14 000
	BR 182620	128	28.575 (1 1/8)	41.275 (1 5/8)	31.750 (1 1/4)	36	1	43 200	65 600	14 000
31.750 (1 1/4)	**BR 202816**	110	31.750 (1 1/4)	44.450 (1 3/4)	25.400 (1)	39.2	1	36 000	53 500	13 000
	BR 202820	138	31.750 (1 1/4)	44.450 (1 3/4)	31.750 (1 1/4)	39.2	1	44 600	70 300	13 000
34.925 (1 3/8)	**BR 223016**	119	34.925 (1 3/8)	47.625 (1 7/8)	25.400 (1)	42.4	1	38 500	60 000	11 000
	BR 223020	149	34.925 (1 3/8)	47.625 (1 7/8)	31.750 (1 1/4)	42.4	1	47 700	78 900	11 000
38.100 (1 1/2)	**BR 243316**	149	38.100 (1 1/2)	52.388 (2 1/16)	25.400 (1)	45.1	1.5	43 700	66 900	11 000
	BR 243320	187	38.100 (1 1/2)	52.388 (2 1/16)	31.750 (1 1/4)	45.1	1.5	54 200	88 200	11 000
41.275 (1 5/8)	**BR 263516**	158	41.275 (1 5/8)	55.562 (2 3/16)	25.400 (1)	48.3	1.5	44 800	70 900	9 500
	BR 263520	199	41.275 (1 5/8)	55.562 (2 3/16)	31.750 (1 1/4)	48.3	1.5	55 600	93 400	9 500
44.450 (1 3/4)	**BR 283716**	170	44.450 (1 3/4)	58.738 (2 5/16)	25.400 (1)	51.5	1.5	47 500	78 200	9 000
	BR 283720	215	44.450 (1 3/4)	58.738 (2 5/16)	31.750 (1 1/4)	51.5	1.5	58 900	103 000	9 000
	BR 283820	250	44.450 (1 3/4)	60.325 (2 3/8)	31.750 (1 1/4)	53.1	1.5	58 900	103 000	9 000
47.625 (1 7/8)	**BR 303920**	225	47.625 (1 7/8)	61.912 (2 7/16)	31.750 (1 1/4)	54.7	1.5	60 100	108 000	8 500

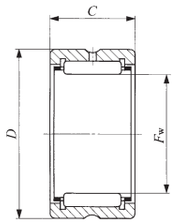
Notes(1) Maximum permissible corner radius of the housing
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

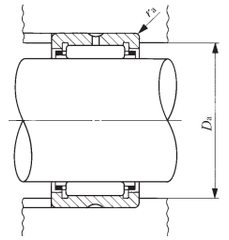
MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring, Inch Series





BR



Shaft dia. 50.800 — 101.600mm

Shaft dia. mm (inch)	Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) F_w	D	C	Standard mounting dimensions mm D_a Max.	$r_{as\,max}$ (1)	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
50.800 (2)	**BR 324116**	190	50.800 (2)	65.088 (2 9/16)	25.400 (1)	57.8	1.5	51 000	89 400	8 000
	BR 324120	240	50.800 (2)	65.088 (2 9/16)	31.750 (1 1/4)	57.8	1.5	63 200	118 000	8 000
57.150 (2 1/4)	**BR 364824**	435	57.150 (2 1/4)	76.200 (3)	38.100 (1 1/2)	69	1.5	90 300	158 000	7 000
	BR 364828	510	57.150 (2 1/4)	76.200 (3)	44.450 (1 3/4)	69	1.5	105 000	191 000	7 000
63.500 (2 1/2)	**BR 405224**	475	63.500 (2 1/2)	82.550 (3 1/4)	38.100 (1 1/2)	74.3	2	94 600	174 000	6 500
	BR 405228	555	63.500 (2 1/2)	82.550 (3 1/4)	44.450 (1 3/4)	74.3	2	110 000	210 000	6 500
69.850 (2 3/4)	**BR 445624**	510	69.850 (2 3/4)	88.900 (3 1/2)	38.100 (1 1/2)	80.7	2	98 700	189 000	5 500
	BR 445628	600	69.850 (2 3/4)	88.900 (3 1/2)	44.450 (1 3/4)	80.7	2	114 000	228 000	5 500
76.200 (3)	**BR 486024**	555	76.200 (3)	95.250 (3 3/4)	38.100 (1 1/2)	87	2	105 000	211 000	5 500
	BR 486028	650	76.200 (3)	95.250 (3 3/4)	44.450 (1 3/4)	87	2	122 000	255 000	5 500
82.550 (3 1/4)	**BR 526828**	990	82.550 (3 1/4)	107.950 (4 1/4)	44.450 (1 3/4)	99.7	2	141 000	259 000	5 000
	BR 526832	1 140	82.550 (3 1/4)	107.950 (4 1/4)	50.800 (2)	99.7	2	154 000	290 000	5 000
88.900 (3 1/2)	**BR 567232**	1 220	88.900 (3 1/2)	114.300 (4 1/2)	50.800 (2)	106.1	2	162 000	316 000	4 500
95.250 (3 3/4)	**BR 607632**	1 290	95.250 (3 3/4)	120.650 (4 3/4)	50.800 (2)	111.4	2.5	169 000	342 000	4 000
101.600 (4)	**BR 648032**	1 370	101.600 (4)	127.000 (5)	50.800 (2)	117.8	2.5	176 000	368 000	4 000

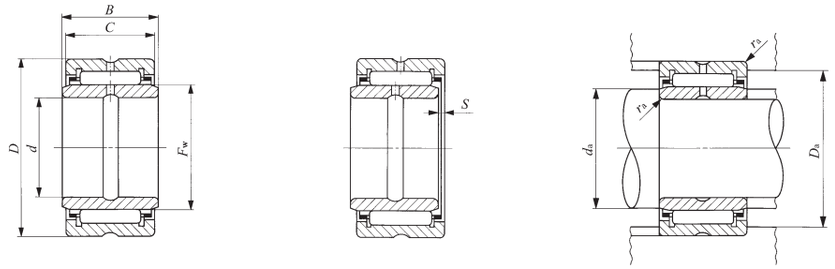
Notes(1) Maximum permissible corner radius of the housing
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. In bearings with a roller set bore diameter F_w of 69.850 mm or less, the outer ring has an oil groove and an oil hole. In others, the outer ring has an oil groove and two oil holes.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring, Inch Series





BRI

Shaft dia. 9.525 — 41.275mm

Shaft dia. mm (inch)	Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) d	D	C	B	F_w	(1) S	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	$r_{as\,max}$ (2)	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(3) rpm	Assembled inner ring
9.525 (3/8)	**BRI 61812**	67.5	9.525 (3/8)	28.575 (1 1/8)	19.050 (3/4)	19.300	15.875 (5/8)	0.3	14	14.5	24.5	0.6	18 900	19 700	25 000	**LRB 61012**
12.700 (1/2)	**BRI 82012**	79.5	12.700 (1/2)	31.750 (1 1/4)	19.050 (3/4)	19.300	19.050 (3/4)	0.3	17.5	18	26.5	1	21 700	24 400	20 000	**LRB 81212**
	BRI 82016	106	12.700 (1/2)	31.750 (1 1/4)	25.400 (1)	25.650	19.050 (3/4)	0.5	17.5	18	26.5	1	27 600	33 100	20 000	**LRB 81216**
15.875 (5/8)	**BRI 102212**	91	15.875 (5/8)	34.925 (1 3/8)	19.050 (3/4)	19.300	22.225 (7/8)	0.3	21	21.2	29.7	1	23 000	27 100	18 000	**LRB 101412**
	BRI 102216	122	15.875 (5/8)	34.925 (1 3/8)	25.400 (1)	25.650	22.225 (7/8)	0.5	21	21.2	29.7	1	29 100	36 800	18 000	**LRB 101416**
19.050 (3/4)	**BRI 122412**	102	19.050 (3/4)	38.100 (1 1/2)	19.050 (3/4)	19.300	25.400 (1)	0.3	24	24.4	32.9	1	25 300	31 900	16 000	**LRB 121612**
	BRI 122416	136	19.050 (3/4)	38.100 (1 1/2)	25.400 (1)	25.650	25.400 (1)	0.5	24	24.4	32.9	1	32 100	43 300	16 000	**LRB 121616**
22.225 (7/8)	**BRI 142616**	152	22.225 (7/8)	41.275 (1 5/8)	25.400 (1)	25.650	28.575 (1 1/8)	0.5	27	27.5	36	1	34 900	49 900	14 000	**LRB 141816**
	BRI 142620	190	22.225 (7/8)	41.275 (1 5/8)	31.750 (1 1/4)	32.000	28.575 (1 1/8)	0.5	27	27.5	36	1	43 200	65 600	14 000	**LRB 141820**
25.400 (1)	**BRI 162816**	166	25.400 (1)	44.450 (1 3/4)	25.400 (1)	25.650	31.750 (1 1/4)	0.5	30.5	30.7	39.2	1	36 000	53 500	13 000	**LRB 162016**
	BRI 162820	210	25.400 (1)	44.450 (1 3/4)	31.750 (1 1/4)	32.000	31.750 (1 1/4)	0.5	30.5	30.7	39.2	1	44 600	70 300	13 000	**LRB 162020**
28.575 (1 1/8)	**BRI 183016**	182	28.575 (1 1/8)	47.625 (1 7/8)	25.400 (1)	25.650	34.925 (1 3/8)	0.5	33.5	33.9	42.4	1	38 500	60 000	11 000	**LRB 182216**
	BRI 183020	225	28.575 (1 1/8)	47.625 (1 7/8)	31.750 (1 1/4)	32.000	34.925 (1 3/8)	0.5	33.5	33.9	42.4	1	47 700	78 900	11 000	**LRB 182220**
31.750 (1 1/4)	**BRI 203316**	220	31.750 (1 1/4)	52.388 (2 1/16)	25.400 (1)	25.650	38.100 (1 1/2)	0.5	37	37.1	45.1	1.5	43 700	66 900	11 000	**LRB 202416**
	BRI 203320	275	31.750 (1 1/4)	52.388 (2 1/16)	31.750 (1 1/4)	32.000	38.100 (1 1/2)	0.5	37	37.1	45.1	1.5	54 200	88 200	11 000	**LRB 202420**
34.925 (1 3/8)	**BRI 223516**	235	34.925 (1 3/8)	55.562 (2 3/16)	25.400 (1)	25.650	41.275 (1 5/8)	0.5	40.2	40.2	48.3	1.5	44 800	70 900	9 500	**LRB 222616**
	BRI 223520	295	34.925 (1 3/8)	55.562 (2 3/16)	31.750 (1 1/4)	32.000	41.275 (1 5/8)	0.5	40.2	40.2	48.3	1.5	55 600	93 400	9 500	**LRB 222620**
38.100 (1 1/2)	**BRI 243716**	250	38.100 (1 1/2)	58.738 (2 5/16)	25.400 (1)	25.650	44.450 (1 3/4)	0.5	43.3	43.4	51.5	1.5	47 500	78 200	9 000	**LRB 242816**
	BRI 243720	315	38.100 (1 1/2)	58.738 (2 5/16)	31.750 (1 1/4)	32.000	44.450 (1 3/4)	0.5	43.3	43.4	51.5	1.5	58 900	103 000	9 000	**LRB 242820**
	BRI 243820	350	38.100 (1 1/2)	60.325 (2 3/8)	31.750 (1 1/4)	32.000	44.450 (1 3/4)	0.5	43.3	43.4	53.1	1.5	58 900	103 000	9 000	**LRB 242820**
	BRI 243920	380	38.100 (1 1/2)	61.912 (2 7/16)	31.750 (1 1/4)	32.000	47.625 (1 7/8)	0.5	43.3	43.4	54.7	1.5	60 100	108 000	8 500	**LRB 243020**
41.275 (1 5/8)	**BRI 264116**	325	41.275 (1 5/8)	65.088 (2 9/16)	25.400 (1)	25.650	50.800 (2)	0.5	48	49	57.8	1.5	51 000	89 400	8 000	**LRB 263216**
	BRI 264120	410	41.275 (1 5/8)	65.088 (2 9/16)	31.750 (1 1/4)	32.000	50.800 (2)	0.5	48	49	57.8	1.5	63 200	118 000	8 000	**LRB 263220**

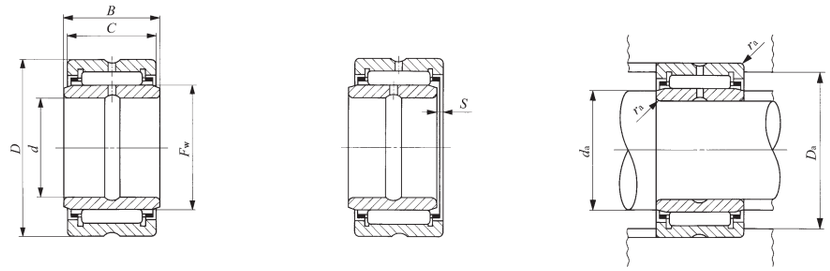
Notes(1) Allowable axial shift amount of inner ring to outer ring
(2) Maximum permissible corner radius of the shaft or housing
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The inner ring and the outer ring each have an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring, Inch Series





BRI

Shaft dia. 44.450 — 88.900mm

Shaft dia. mm (inch)	Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) d	D	C	B	F_w	S(1)	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	$r_{as\,max}$(2)	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(3) rpm	Assembled inner ring
44.450 (1¾)	**BRI 284824**	735	44.450 (1¾)	76.200 (3)	38.100 (1½)	38.350	57.150 (2¼)	1	52.5	55	69	1.5	90 300	158 000	7 000	**LRB 283624**
	BRI 284828	855	44.450 (1¾)	76.200 (3)	44.450 (1¾)	44.700	57.150 (2¼)	1	52.5	55	69	1.5	105 000	191 000	7 000	**LRB 283628**
50.800 (2)	**BRI 325224**	810	50.800 (2)	82.550 (3¼)	38.100 (1½)	38.350	63.500 (2½)	1	58	61	74.3	2	94 600	174 000	6 500	**LRB 324024**
	BRI 325228	945	50.800 (2)	82.550 (3¼)	44.450 (1¾)	44.700	63.500 (2½)	1	58	61	74.3	2	110 000	210 000	6 500	**LRB 324028**
57.150 (2¼)	**BRI 365624**	885	57.150 (2¼)	88.900 (3½)	38.100 (1½)	38.350	69.850 (2¾)	1	65	67	80.7	2	98 700	189 000	5 500	**LRB 364424**
	BRI 365628	1 040	57.150 (2¼)	88.900 (3½)	44.450 (1¾)	44.700	69.850 (2¾)	1	65	67	80.7	2	114 000	228 000	5 500	**LRB 364428**
63.500 (2½)	**BRI 406024**	965	63.500 (2½)	95.250 (3¾)	38.100 (1½)	38.350	76.200 (3)	1	71	73	87	2	105 000	211 000	5 500	**LRB 404824**
	BRI 406028	1 130	63.500 (2½)	95.250 (3¾)	44.450 (1¾)	44.700	76.200 (3)	1	71	73	87	2	122 000	255 000	5 500	**LRB 404828**
69.850 (2¾)	**BRI 446828**	1 520	69.850 (2¾)	107.950 (4¼)	44.450 (1¾)	44.700	82.550 (3¼)	1.5	77	79	99.7	2	141 000	259 000	5 000	**LRB 445228**
	BRI 446832	1 740	69.850 (2¾)	107.950 (4¼)	50.800 (2)	51.050	82.550 (3¼)	3	77	79	99.7	2	154 000	290 000	5 000	**LRB 445232**
76.200 (3)	**BRI 487232**	1 860	76.200 (3)	114.300 (4½)	50.800 (2)	51.050	88.900 (3½)	3	83.5	86	106.1	2	162 000	316 000	4 500	**LRB 485632**
82.550 (3¼)	**BRI 527632**	1 980	82.550 (3¼)	120.650 (4¾)	50.800 (2)	51.050	95.250 (3¾)	3	91	93	111.4	2.5	169 000	342 000	4 000	**LRB 526032**
88.900 (3½)	**BRI 568032**	2 120	88.900 (3½)	127.000 (5)	50.800 (2)	51.050	101.600 (4)	3	97	99	117.8	2.5	176 000	368 000	4 000	**LRB 566432**

Notes(1) Allowable axial shift amount of inner ring to outer ring
(2) Maximum permissible corner radius of the shaft or housing
(3) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.

Remarks1. In bearings with a bearing bore diameter, d, of 57.150 mm or less, the outer ring has an oil groove and an oil hole. In bearings with a bearing bore diameter, d, of 76.200 mm or less, the inner ring has an oil groove and an oil hole. In others, the inner ring and the outer ring each have an oil groove and two oil holes.
2. No grease is prepacked. Perform proper lubrication.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

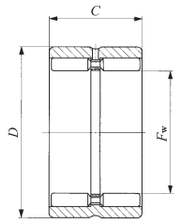
Without Inner Ring, Inch Series



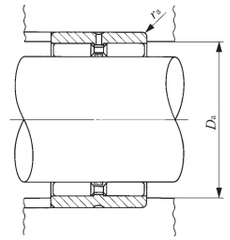
Shaft dia. 15.875 — 50.800mm

Shaft dia. mm (inch)	Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) F_w	D	C	Standard mounting dimensions mm D_a Max.	$r_{as\,max}$ (1)	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
15.875 (5/8)	**GBR 101812**	55.5	15.875 (5/8)	28.575 (1 1/8)	19.050 (3/4)	24.5	0.6	23 500	28 500	9 500
19.050 (3/4)	**GBR 122012**	63	19.050 (3/4)	31.750 (1 1/4)	19.050 (3/4)	27	0.6	26 400	34 500	8 000
22.225 (7/8)	**GBR 142212**	71	22.225 (7/8)	34.925 (1 3/8)	19.050 (3/4)	30	0.6	28 600	40 100	7 000
	GBR 142216	95.5	22.225 (7/8)	34.925 (1 3/8)	25.400 (1)	30	0.6	38 300	58 300	7 000
25.400 (1)	**GBR 162412**	79	25.400 (1)	38.100 (1 1/2)	19.050 (3/4)	33.3	0.6	31 000	46 100	6 000
	GBR 162416	106	25.400 (1)	38.100 (1 1/2)	25.400 (1)	33.3	0.6	41 400	67 100	6 000
28.575 (1 1/8)	**GBR 182616**	117	28.575 (1 1/8)	41.275 (1 5/8)	25.400 (1)	36.3	0.6	43 900	75 300	5 500
31.750 (1 1/4)	**GBR 202816**	128	31.750 (1 1/4)	44.450 (1 3/4)	25.400 (1)	39.6	0.6	46 600	83 900	4 500
34.925 (1 3/8)	**GBR 223016**	137	34.925 (1 3/8)	47.625 (1 7/8)	25.400 (1)	42.8	0.6	49 500	91 800	4 500
38.100 (1 1/2)	**GBR 243316**	168	38.100 (1 1/2)	52.388 (2 1/16)	25.400 (1)	47.3	0.6	54 200	97 700	4 000
	GBR 243320	205	38.100 (1 1/2)	52.388 (2 1/16)	31.750 (1 1/4)	47.3	0.6	64 100	121 000	4 000
41.275 (1 5/8)	**GBR 263516**	180	41.275 (1 5/8)	55.562 (2 3/16)	25.400 (1)	50.5	0.6	56 600	105 000	3 500
	GBR 263520	220	41.275 (1 5/8)	55.562 (2 3/16)	31.750 (1 1/4)	50.5	0.6	67 000	130 000	3 500
44.450 (1 3/4)	**GBR 283720**	235	44.450 (1 3/4)	58.738 (2 5/16)	31.750 (1 1/4)	53.7	0.6	69 700	141 000	3 500
	GBR 283820	275	44.450 (1 3/4)	60.325 (2 3/8)	31.750 (1 1/4)	55.3	0.6	69 700	141 000	3 500
47.625 (1 7/8)	**GBR 303920**	250	47.625 (1 7/8)	61.912 (2 7/16)	31.750 (1 1/4)	56.2	1	72 400	150 000	3 000
50.800 (2)	**GBR 324116**	215	50.800 (2)	65.088 (2 9/16)	25.400 (1)	59.2	1	63 100	130 000	3 000
	GBR 324120	265	50.800 (2)	65.088 (2 9/16)	31.750 (1 1/4)	59.2	1	74 600	162 000	3 000

Notes(1) Maximum permissible corner radius of the housing
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.



GBR

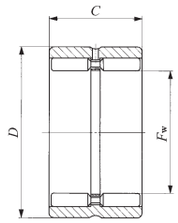


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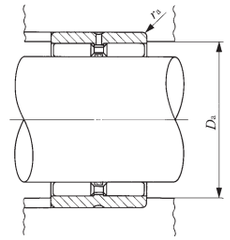
MACHINED TYPE NEEDLE ROLLER BEARINGS

Without Inner Ring, Inch Series





GBR



Shaft dia. 57.150 — 107.950mm

Shaft dia. mm (inch)	Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) F_w	D	C	Standard mounting dimensions mm D_a Max.	$r_{as\,max}$ (1)	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
57.150 (2 1/4)	**GBR 364824**	490	57.150 (2 1/4)	76.200 (3)	38.100 (1 1/2)	69.2	1.5	113 000	224 000	2 500
	GBR 364828	580	57.150 (2 1/4)	76.200 (3)	44.450 (1 3/4)	69.2	1.5	133 000	276 000	2 500
63.500 (2 1/2)	**GBR 405224**	535	63.500 (2 1/2)	82.550 (3 1/4)	38.100 (1 1/2)	75.7	1.5	120 000	248 000	2 500
	GBR 405228	635	63.500 (2 1/2)	82.550 (3 1/4)	44.450 (1 3/4)	75.7	1.5	141 000	306 000	2 500
69.850 (2 3/4)	**GBR 445624**	585	69.850 (2 3/4)	88.900 (3 1/2)	38.100 (1 1/2)	82	1.5	125 000	273 000	2 000
	GBR 445628	690	69.850 (2 3/4)	88.900 (3 1/2)	44.450 (1 3/4)	82	1.5	147 000	336 000	2 000
76.200 (3)	**GBR 486024**	630	76.200 (3)	95.250 (3 3/4)	38.100 (1 1/2)	88	1.5	131 000	298 000	2 000
	GBR 486028	745	76.200 (3)	95.250 (3 3/4)	44.450 (1 3/4)	88	1.5	154 000	368 000	2 000
82.550 (3 1/4)	**GBR 526828**	1 100	82.550 (3 1/4)	107.950 (4 1/4)	44.450 (1 3/4)	99.9	1.5	193 000	396 000	1 800
	GBR 526832	1 240	82.550 (3 1/4)	107.950 (4 1/4)	50.800 (2)	99.9	1.5	214 000	452 000	1 800
88.900 (3 1/2)	**GBR 567232**	1 330	88.900 (3 1/2)	114.300 (4 1/2)	50.800 (2)	106.3	1.5	221 000	488 000	1 700
95.250 (3 3/4)	**GBR 607632**	1 420	95.250 (3 3/4)	120.650 (4 3/4)	50.800 (2)	112.6	1.5	228 000	522 000	1 600
101.600 (4)	**GBR 648032**	1 500	101.600 (4)	127.000 (5)	50.800 (2)	119	1.5	237 000	556 000	1 500
107.950 (4 1/4)	**GBR 688432**	1 580	107.950 (4 1/4)	133.350 (5 1/4)	50.800 (2)	125.3	1.5	242 000	590 000	1 400

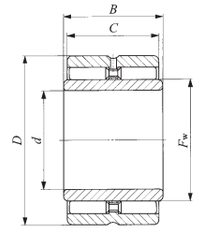
Notes(1) Maximum permissible corner radius of the housing
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

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1mm=0.03937inch

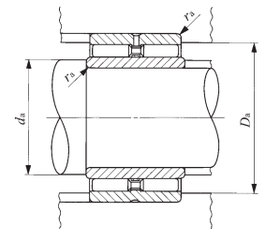
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring, Inch Series





GBRI



Shaft dia. 9.525 — 41.275mm

Shaft dia. mm (inch)	Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) d	D	C	B	F_w	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	$r_{as\,max}$ (1)	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (2) rpm	Assembled inner ring
9.525 (3/8)	**GBRI 61812**	74	9.525 (3/8)	28.575 (1 1/8)	19.050 (3/4)	19.300	15.875 (5/8)	14	14.5	24.5	0.6	23 500	28 500	9 500	**LRBZ 61012**
12.700 (1/2)	**GBRI 82012**	86.5	12.700 (1/2)	31.750 (1 1/4)	19.050 (3/4)	19.300	19.050 (3/4)	17.5	18	27	0.6	26 400	34 500	8 000	**LRBZ 81212**
15.875 (5/8)	**GBRI 102212**	99	15.875 (5/8)	34.925 (1 3/8)	19.050 (3/4)	19.300	22.225 (7/8)	21	21.2	30	0.6	28 600	40 100	7 000	**LRBZ 101412**
	GBRI 102216	133	15.875 (5/8)	34.925 (1 3/8)	25.400 (1)	25.650	22.225 (7/8)	21	21.2	30	0.6	38 300	58 300	7 000	**LRBZ 101416**
19.050 (3/4)	**GBRI 122412**	112	19.050 (3/4)	38.100 (1 1/2)	19.050 (3/4)	19.300	25.400 (1)	24	24.4	33.3	0.6	31 000	46 100	6 000	**LRBZ 121612**
	GBRI 122416	150	19.050 (3/4)	38.100 (1 1/2)	25.400 (1)	25.650	25.400 (1)	24	24.4	33.3	0.6	41 400	67 100	6 000	**LRBZ 121616**
22.225 (7/8)	**GBRI 142616**	167	22.225 (7/8)	41.275 (1 5/8)	25.400 (1)	25.650	28.575 (1 1/8)	27	27.5	36.3	0.6	43 900	75 300	5 500	**LRBZ 141816**
25.400 (1)	**GBRI 162816**	184	25.400 (1)	44.450 (1 3/4)	25.400 (1)	25.650	31.750 (1 1/4)	30.5	30.7	39.6	0.6	46 600	83 900	4 500	**LRBZ 162016**
28.575 (1 1/8)	**GBRI 183016**	200	28.575 (1 1/8)	47.625 (1 7/8)	25.400 (1)	25.650	34.925 (1 3/8)	33.5	33.9	42.8	0.6	49 500	91 800	4 500	**LRBZ 182216**
31.750 (1 1/4)	**GBRI 203316**	235	31.750 (1 1/4)	52.388 (2 1/16)	25.400 (1)	25.650	38.100 (1 1/2)	37	37.1	47.3	0.6	54 200	97 700	4 000	**LRBZ 202416**
	GBRI 203320	291	31.750 (1 1/4)	52.388 (2 1/16)	31.750 (1 1/4)	32.000	38.100 (1 1/2)	37	37.1	47.3	0.6	64 100	121 000	4 000	**LRBZ 202420**
34.925 (1 3/8)	**GBRI 223516**	255	34.925 (1 3/8)	55.562 (2 3/16)	25.400 (1)	25.650	41.275 (1 5/8)	40.2	40.2	50.5	0.6	56 600	105 000	3 500	**LRBZ 222616**
	GBRI 223520	316	34.925 (1 3/8)	55.562 (2 3/16)	31.750 (1 1/4)	32.000	41.275 (1 5/8)	40.2	40.2	50.5	0.6	67 000	130 000	3 500	**LRBZ 222620**
38.100 (1 1/2)	**GBRI 243720**	335	38.100 (1 1/2)	58.738 (2 5/16)	31.750 (1 1/4)	32.000	44.450 (1 3/4)	43.3	43.4	53.7	0.6	69 700	141 000	3 500	**LRBZ 242820**
	GBRI 243820	375	38.100 (1 1/2)	60.325 (2 3/8)	31.750 (1 1/4)	32.000	44.450 (1 3/4)	43.3	43.4	55.3	0.6	69 700	141 000	3 500	**LRBZ 242820**
	GBRI 243920	410	38.100 (1 1/2)	61.912 (2 7/16)	31.750 (1 1/4)	32.000	47.625 (1 7/8)	43.3	45	56.2	1	72 400	150 000	3 000	**LRBZ 243020**
41.275 (1 5/8)	**GBRI 264116**	350	41.275 (1 5/8)	65.088 (2 9/16)	25.400 (1)	25.650	50.800 (2)	48	49	59.2	1	63 100	130 000	3 000	**LRBZ 263216**
	GBRI 264120	435	41.275 (1 5/8)	65.088 (2 9/16)	31.750 (1 1/4)	32.000	50.800 (2)	48	49	59.2	1	74 600	162 000	3 000	**LRBZ 263220**

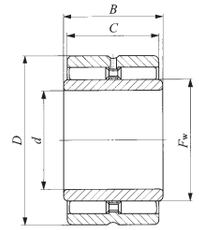
Notes(1) Maximum permissible corner radius of the shaft or housing
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

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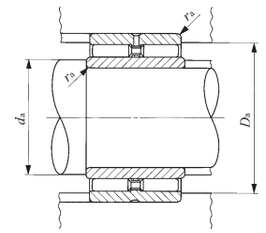
MACHINED TYPE NEEDLE ROLLER BEARINGS

With Inner Ring, Inch Series





GBRI



Shaft dia. 44.450 — 95.250mm

Shaft dia. mm (inch)	Identification number	Mass (Ref.) g	Boundary dimensions mm(inch) d	D	C	B	F_w	Standard mounting dimensions mm d_a Min.	d_a Max.	D_a Max.	$r_{as\,max}$ (1)	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed (2) rpm	Assembled inner ring
44.450 (1¾)	**GBRI 284824**	790	44.450 (1¾)	76.200 (3)	38.100 (1½)	38.350	57.150 (2¼)	52.5	55	69.2	1.5	113 000	224 000	2 500	**LRBZ 283624**
	GBRI 284828	925	44.450 (1¾)	76.200 (3)	44.450 (1¾)	44.700	57.150 (2¼)	52.5	55	69.2	1.5	133 000	276 000	2 500	**LRBZ 283628**
50.800 (2)	**GBRI 325224**	870	50.800 (2)	82.550 (3¼)	38.100 (1½)	38.350	63.500 (2½)	58	61	75.7	1.5	120 000	248 000	2 500	**LRBZ 324024**
	GBRI 325228	1 030	50.800 (2)	82.550 (3¼)	44.450 (1¾)	44.700	63.500 (2½)	58	61	75.7	1.5	141 000	306 000	2 500	**LRBZ 324028**
57.150 (2¼)	**GBRI 365624**	955	57.150 (2¼)	88.900 (3½)	38.100 (1½)	38.350	69.850 (2¾)	65	67	82	1.5	125 000	273 000	2 000	**LRBZ 364424**
	GBRI 365628	1 130	57.150 (2¼)	88.900 (3½)	44.450 (1¾)	44.700	69.850 (2¾)	65	67	82	1.5	147 000	336 000	2 000	**LRBZ 364428**
63.500 (2½)	**GBRI 406024**	1 040	63.500 (2½)	95.250 (3¾)	38.100 (1½)	38.350	76.200 (3)	71	73	88	1.5	131 000	298 000	2 000	**LRBZ 404824**
	GBRI 406028	1 230	63.500 (2½)	95.250 (3¾)	44.450 (1¾)	44.700	76.200 (3)	71	73	88	1.5	154 000	368 000	2 000	**LRBZ 404828**
69.850 (2¾)	**GBRI 446828**	1 630	69.850 (2¾)	107.950 (4¼)	44.450 (1¾)	44.700	82.550 (3¼)	77	79	99.9	1.5	193 000	396 000	1 800	**LRBZ 445228**
	GBRI 446832	1 840	69.850 (2¾)	107.950 (4¼)	50.800 (2)	51.050	82.550 (3¼)	77	79	99.9	1.5	214 000	452 000	1 800	**LRBZ 445232**
76.200 (3)	**GBRI 487232**	1 970	76.200 (3)	114.300 (4½)	50.800 (2)	51.050	88.900 (3½)	83.5	86	106.3	1.5	221 000	488 000	1 700	**LRBZ 485632**
82.550 (3¼)	**GBRI 527632**	2 110	82.550 (3¼)	120.650 (4¾)	50.800 (2)	51.050	95.250 (3¾)	91	93	112.6	1.5	228 000	522 000	1 600	**LRBZ 526032**
88.900 (3½)	**GBRI 568032**	2 250	88.900 (3½)	127.000 (5)	50.800 (2)	51.050	101.600 (4)	97	99	119	1.5	237 000	556 000	1 500	**LRBZ 566432**
95.250 (3¾)	**GBRI 608432**	2 380	95.250 (3¾)	133.350 (5¼)	50.800 (2)	51.050	107.950 (4¼)	103	105	125.3	1.5	242 000	590 000	1 400	**LRBZ 606832**

Notes(1) Maximum permissible corner radius of the shaft or housing
(2) Allowable rotational speed applies to oil lubrication. For grease lubrication, a maximum of 60% of this value is allowable.
Remarks1. The outer ring has an oil groove and an oil hole.
2. No grease is prepacked. Perform proper lubrication.

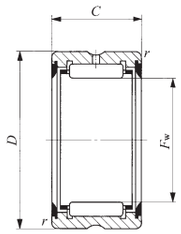
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MACHINED TYPE NEEDLE ROLLER BEARINGS

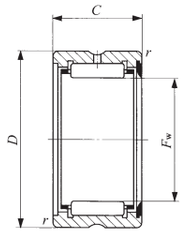
With Seal, Without Inner Ring



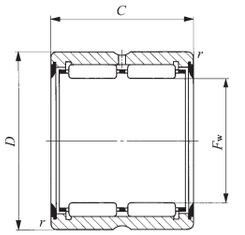




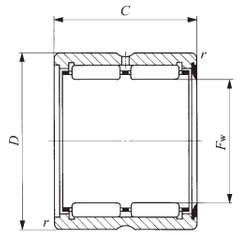
RNA49···UU
RNA69···UU($F_w \leqq 35$)



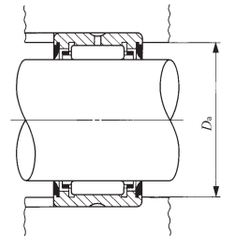
RNA49···U
RNA69···U($F_w \leqq 35$)



RNA69···UU



RNA69···U



Shaft dia. 14 — 45mm

Shaft dia. mm	Identification number: With two seals	With one seal	With two seals	With one seal	Mass (Ref.) g	Boundary dimensions mm: F_w	D	C	$r_{s\,min}$ (1)
14	**RNA 4900UU**	**RNA 4900U**	—	—	16.3	14	22	13	0.3
16	**RNA 4901UU**	**RNA 4901U**	—	—	17.9	16	24	13	0.3
	—	—	**RNA 6901UU**	**RNA 6901U**	30	16	24	22	0.3
18	**RNA 49/14UU**	**RNA 49/14U**	—	—	19.7	18	26	13	0.3
20	**RNA 4902UU**	**RNA 4902U**	—	—	21.5	20	28	13	0.3
	—	—	**RNA 6902UU**	**RNA 6902U**	37.5	20	28	23	0.3
22	**RNA 4903UU**	**RNA 4903U**	—	—	23	22	30	13	0.3
	—	—	**RNA 6903UU**	**RNA 6903U**	40.5	22	30	23	0.3
25	**RNA 4904UU**	**RNA 4904U**	—	—	54.5	25	37	17	0.3
	—	—	**RNA 6904UU**	**RNA 6904U**	95.5	25	37	30	0.3
28	**RNA 49/22UU**	**RNA 49/22U**	—	—	55.5	28	39	17	0.3
	—	—	**RNA 69/22UU**	**RNA 69/22U**	97.5	28	39	30	0.3
30	**RNA 4905UU**	**RNA 4905U**	—	—	63	30	42	17	0.3
	—	—	**RNA 6905UU**	**RNA 6905U**	111	30	42	30	0.3
32	**RNA 49/28UU**	**RNA 49/28U**	—	—	75.5	32	45	17	0.3
	—	—	**RNA 69/28UU**	**RNA 69/28U**	133	32	45	30	0.3
35	**RNA 4906UU**	**RNA 4906U**	—	—	71	35	47	17	0.3
	—	—	**RNA 6906UU**	**RNA 6906U**	125	35	47	30	0.3
40	**RNA 49/32UU**	**RNA 49/32U**	—	—	94.5	40	52	20	0.6
	—	—	**RNA 69/32UU**	**RNA 69/32U**	170	40	52	36	0.6
42	**RNA 4907UU**	**RNA 4907U**	—	—	112	42	55	20	0.6
	—	—	**RNA 6907UU**	**RNA 6907U**	200	42	55	36	0.6
45	**RNA 49/38UU**	**RNA 49/38U**	—	—	119	45	58	20	0.6

Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
20	8 080	8 490	14 000
22	8 470	9 320	12 000
22	15 500	20 400	12 000
24	9 260	10 800	11 000
26	9 570	11 600	9 500
26	18 500	27 100	9 500
28	10 300	13 100	8 500
28	19 800	30 600	8 500
35	18 000	20 500	7 500
35	33 000	44 600	7 500
37	18 300	23 700	7 000
37	33 800	52 000	7 000
40	20 300	25 100	6 500
40	39 200	58 700	6 500
43	21 000	26 800	6 000
43	38 900	59 100	6 000
45	21 500	28 400	5 500
45	40 100	63 000	5 500
48	29 400	44 200	5 000
48	50 300	88 300	5 000
51	30 100	46 300	4 500
51	51 600	92 600	4 500
54	31 600	50 400	4 000

Notes(1) Minimum allowable value of chamfer dimension r
(2) Allowable rotational speed applies to grease lubrication.
Remarks1. The outer ring has an oil groove and an oil hole.
2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

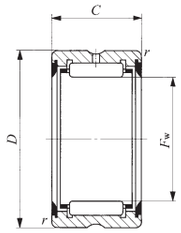
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MACHINED TYPE NEEDLE ROLLER BEARINGS

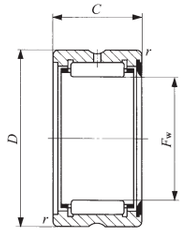
With Seal, Without Inner Ring



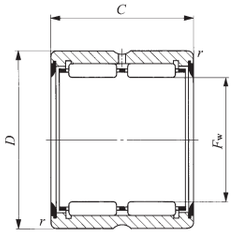




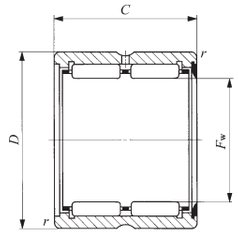
RNA49···UU



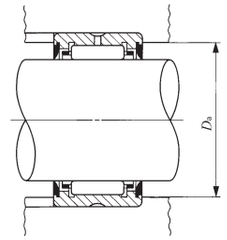
RNA49···U



RNA69···UU



RNA69···U



Shaft dia. 48 — 85mm

Shaft dia. mm	Identification number: With two seals	With one seal	With two seals	With one seal	Mass (Ref.) g	Boundary dimensions mm: F_w	D	C	$r_{s\,min}$ (1)
48	**RNA 4908UU**	**RNA 4908U**	—	—	150	48	62	22	0.6
	—	—	**RNA 6908UU**	**RNA 6908U**	270	48	62	40	0.6
50	**RNA 49/42UU**	**RNA 49/42U**	—	—	173	50	65	22	0.6
52	**RNA 4909UU**	**RNA 4909U**	—	—	197	52	68	22	0.6
	—	—	**RNA 6909UU**	**RNA 6909U**	355	52	68	40	0.6
55	**RNA 49/48UU**	**RNA 49/48U**	—	—	187	55	70	22	0.6
58	**RNA 4910UU**	**RNA 4910U**	—	—	177	58	72	22	0.6
	—	—	**RNA 6910UU**	**RNA 6910U**	320	58	72	40	0.6
60	**RNA 49/52UU**	**RNA 49/52U**	—	—	200	60	75	22	0.6
63	**RNA 4911UU**	**RNA 4911U**	—	—	265	63	80	25	1
	—	—	**RNA 6911UU**	**RNA 6911U**	470	63	80	45	1
65	**RNA 49/58UU**	**RNA 49/58U**	—	—	275	65	82	25	1
68	**RNA 4912UU**	**RNA 4912U**	—	—	285	68	85	25	1
	—	—	**RNA 6912UU**	**RNA 6912U**	505	68	85	45	1
70	**RNA 49/62UU**	**RNA 49/62U**	—	—	320	70	88	25	1
72	**RNA 4913UU**	**RNA 4913U**	—	—	325	72	90	25	1
	—	—	**RNA 6913UU**	**RNA 6913U**	580	72	90	45	1
75	**RNA 49/68UU**	**RNA 49/68U**	—	—	465	75	95	30	1
80	**RNA 4914UU**	**RNA 4914U**	—	—	495	80	100	30	1
	—	—	**RNA 6914UU**	**RNA 6914U**	910	80	100	54	1
85	**RNA 4915UU**	**RNA 4915U**	—	—	520	85	105	30	1
	—	—	**RNA 6915UU**	**RNA 6915U**	960	85	105	54	1

Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
58	37 200	58 400	4 000
58	63 700	117 000	4 000
61	38 000	60 900	4 000
64	38 900	63 400	3 500
64	66 600	127 000	3 500
66	39 600	66 100	3 500
68	41 300	71 100	3 500
68	70 800	142 000	3 500
71	42 100	73 600	3 000
75	52 200	85 700	3 000
75	89 400	171 000	3 000
77	53 400	89 200	3 000
80	54 500	92 800	3 000
80	93 400	186 000	3 000
83	55 700	96 300	2 500
85	56 800	99 800	2 500
85	97 400	200 000	2 500
90	73 900	133 000	2 500
95	76 900	143 000	2 500
95	124 000	281 000	2 500
100	79 600	153 000	2 000
100	128 000	299 000	2 000

Notes(1) Minimum allowable value of chamfer dimension r

(2) Allowable rotational speed applies to grease lubrication.

Remarks1. The outer ring has an oil groove and an oil hole.

2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

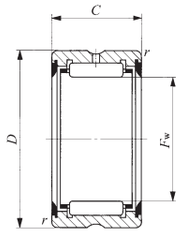
1N=0.102kgf=0.2248lbs.
1mm=0.03937inch

MACHINED TYPE NEEDLE ROLLER BEARINGS

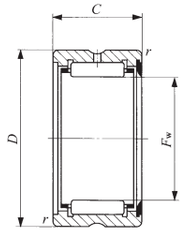
With Seal, Without Inner Ring



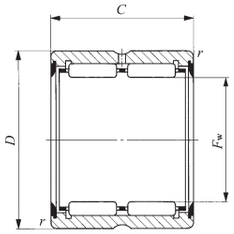




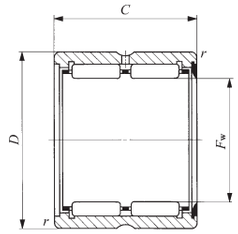
RNA49···UU



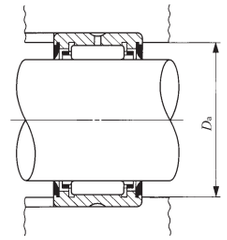
RNA49···U



RNA69···UU



RNA69···U



Shaft dia. 90 — 160mm

Shaft dia. mm	Identification number: With two seals	With one seal	With two seals	With one seal	Mass (Ref.) g	Boundary dimensions mm: F_w	D	C	$r_{s\,min}$ (1)
90	**RNA 4916UU**	**RNA 4916U**	—	—	545	90	110	30	1
	—	—	**RNA 6916UU**	**RNA 6916U**	1 010	90	110	54	1
95	**RNA 49/82UU**	**RNA 49/82U**	—	—	570	95	115	30	1
100	**RNA 4917UU**	**RNA 4917U**	—	—	695	100	120	35	1.1
	—	—	**RNA 6917UU**	**RNA 6917U**	1 300	100	120	63	1.1
105	**RNA 4918UU**	**RNA 4918U**	—	—	730	105	125	35	1.1
	—	—	**RNA 6918UU**	**RNA 6918U**	1 360	105	125	63	1.1
110	**RNA 4919UU**	**RNA 4919U**	—	—	760	110	130	35	1.1
	—	—	**RNA 6919UU**	**RNA 6919U**	1 420	110	130	63	1.1
115	**RNA 4920UU**	**RNA 4920U**	—	—	1 200	115	140	40	1.1
125	**RNA 4922UU**	**RNA 4922U**	—	—	1 280	125	150	40	1.1
135	**RNA 4924UU**	**RNA 4924U**	—	—	1 940	135	165	45	1.1
150	**RNA 4926UU**	**RNA 4926U**	—	—	2 360	150	180	50	1.5
160	**RNA 4928UU**	**RNA 4928U**	—	—	2 510	160	190	50	1.5

Standard mounting dimension D_a Max. mm	Basic dynamic load rating C N	Basic static load rating C_0 N	Allowable rotational speed(2) rpm
105	80 700	158 000	2 000
105	132 000	317 000	2 000
110	83 200	168 000	2 000
113.5	103 000	225 000	1 900
113.5	168 000	448 000	1 900
118.5	106 000	238 000	1 800
118.5	172 000	471 000	1 800
123.5	109 000	250 000	1 700
123.5	177 000	493 000	1 700
133.5	134 000	297 000	1 700
143.5	140 000	322 000	1 500
158.5	178 000	410 000	1 400
172	206 000	511 000	1 300
182	214 000	549 000	1 200

Notes(1) Minimum allowable value of chamfer dimension r

(2) Allowable rotational speed applies to grease lubrication.

Remarks1. The outer ring has an oil groove and an oil hole.

2. Bearings with seals on both sides are provided with prepacked grease. Bearings with a seal on one side are not provided with prepacked grease. Perform proper lubrication for use.

1N=0.102kgf=0.2248lbs.
1mm=0.03937inch